Causal LSTM for RUL Prediction On CMAPSS Dataset

Bonafide record of work done by

Abirami M (22z204)

Manojkumar (22z236)

Monish Rajan L (22z240)

O Keerthi (22z243)

Kapil Arif K (22z432)

GUIDE: Dr.Anisha C.D

Dissertation submitted in the partial fulfillment of the requirements for the degree of

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Dr. Anisha C.D	**Dr. G.Sudha Sadasivam**
Faculty guide	HoD of the department

Certified that the candidate was examined in the viva-voce examination held on ……

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(Internal Examiner)	(External Examiner)

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SYNOPSIS

This project aims to predict the Remaining Useful Life (RUL) of aerospace engines, which is a critical task in Prognostics and Health Management (PHM) to prevent unexpected failures and enable proactive maintenance decisions. Traditional deep learning models such as LSTM and CNN effectively capture temporal patterns but often fail to consider causal relationships, leading to limited interpretability and suboptimal predictions.

To overcome these challenges, this project compares three different architectures for RUL prediction - a baseline LSTM, a Causal Representation Learning (CRL) autoencoder, and an LSTM model trained on features extracted through CRL. The project utilizes the NASA CMAPSS dataset, a widely recognized benchmark dataset that contains simulated data under varying operational conditions and fault modes.The experimental results demonstrate that incorporating causal structure significantly enhances prediction accuracy and supports more informed maintenance decision-making.

CHAPTER 1
INTRODUCTION

1.1 Problem Identification

Ensuring the safety and continuous operation of aircraft engines is one of the major challenges in the aerospace industry. Unexpected failures can cause major losses and serious safety risks.To handle this issue, modern maintenance systems focus on predicting possible failures before they occur. This approach, known as Prognostics and Health Management (PHM), estimates the Remaining Useful Life (RUL) of components, which represents the number of operating cycles left before failure. Predicting RUL accurately helps in planning maintenance schedules, reducing unplanned downtime, and improving overall system reliability.

1.2 Problem Statement

With advancements in sensing and data collection technologies, large amounts of engine health data are available for analysis. Data-driven techniques, especially Long Short-Term Memory (LSTM) networks, are widely used to learn degradation patterns from this sensor data. However, these models mainly capture correlations between features and do not always identify the true cause of degradation. As a result, their performance may drop when tested on new fault modes or operating conditions. The NASA C-MAPSS dataset provides realistic turbofan engine data with multiple fault modes and varying operating environments. It is therefore chosen in this project as a benchmark for building and evaluating models for RUL prediction.

1.3 Motivation of the Work

The main motivation of this project is to make RUL prediction models both accurate and reliable. Traditional deep learning models provide good accuracy but act as black boxes, offering little understanding of why certain predictions are made. This lack of transparency makes it difficult to trust their predictions in critical aerospace applications. In this project, causal learning techniques are introduced to extract more meaningful information from the data. By combining Causal Representation Learning (CRL) with LSTM, the model focuses on the actual causes of degradation rather than spurious patterns. The project compares three architectures-Baseline LSTM, CRL Autoencoder, and CRL-based LSTM to identify which provides the best balance between accuracy and interpretability.

This report is organised as follows-Chapter 1 includes the introduction,the problem statement and motivation,Chapter 2 includes the literature survey of the domain and various models available in the RUL prediction,Chapter 3 talks about the methodology used and followed by the evaluation metrics and result analysis in Chapter 4 and finally Chapter 5 includes the conclusion and future works of this project.

CHAPTER 2

LITERATURE SURVEY

2.1. Maintenance and Remaining Useful Life in Aerospace

The modern aerospace sector is steadily shifting from rigid calendar based maintenance schedules toward intelligent, Condition Based Maintenance (CBM). In this approach, service decisions depend on the actual condition of a component, determined through continuous monitoring and real time sensor data, instead of fixed inspection intervals. At the core of this transformation is the concept of Remaining Useful Life (RUL), which estimates how long a component can continue to operate safely before it requires attention or replacement. Reliable RUL predictions allow organizations to plan maintenance more efficiently, reducing unnecessary downtime and costs while maintaining safety standards (Ferreira et al., 2022).

A key milestone in this area was the introduction of NASA's C MAPSS (Commercial Modular Aero Propulsion System Simulation) dataset and the PHM08 challenge (Saxena & Goebel, 2008). These provided researchers with a realistic and controlled environment to study the degradation of turbofan engines. The simulator generates detailed sensor readings that capture multiple operational conditions and fault types, offering an ideal benchmark for developing and validating new RUL prediction models (Ramasso, 2014).

2.2. Classical Machine Learning Approaches and Feature Engineering

Before the rise of deep learning, RUL estimation largely relied on classical machine learning models supported by handcrafted features. Researchers extracted meaningful indicators from sensor signals such as statistical trends, frequency information, and degradation profiles to describe equipment health. Algorithms including Random Forests (Wu et al., 2017), Gradient Boosting Machines, and Support Vector Regression (SVR) became popular because of their interpretability and strong predictive power (Lei et al., 2018). When combined with domain specific features, these models delivered dependable results, particularly in smaller or safety critical datasets where deep models can overfit. Recent studies confirm that such interpretable methods continue to be valuable in practical maintenance applications (Ensarioğlu et al., 2023).

2.3. Deep Learning Models and Modern Architectures

The introduction of deep learning revolutionized RUL modeling by allowing neural networks to learn directly from raw sensor data. These models can capture complex short term and long term dependencies within multivariate time series, offering a more comprehensive view of how components degrade over time.

2.3.1. Long Short Term Memory Networks

Long Short Term Memory (LSTM) networks were among the first deep learning models to show consistent promise for RUL prediction. Their ability to retain long range temporal information made them effective for analyzing sequential sensor data (Ramasso, 2014). Zheng et al. (2017) demonstrated their potential on the C MAPSS dataset by introducing a piecewise RUL capping method that reduced noise during the early operational cycles. This established LSTMs as a reliable foundation for time dependent prognostics. Later

enhancements such as bidirectional layers and attention mechanisms further improved their capacity to capture contextual relationships.

2.3.2. Convolutional and Hybrid CNN RNN Frameworks

Researchers soon explored combining Convolutional Neural Networks (CNNs) with recurrent layers such as LSTMs or GRUs to leverage both spatial and temporal features. In a typical configuration, CNNs first extract compact representations from sliding sensor windows, which are then passed to recurrent layers for sequence learning (Solis Martín et al., 2021). This structure allows CNNs to detect local patterns or anomalies, while RNNs model the broader degradation trajectory (Ensarioğlu et al., 2023).

2.3.3. Attention and Transformer Architectures

In recent years, Attention mechanisms and Transformer architectures have advanced the field by overcoming the limitations of traditional recurrent models. Unlike RNNs, Transformers can directly capture long range dependencies without suffering from vanishing gradients. The Dual Aspect Self Attention Transformer (DAST) proposed by Zhang et al. (2021) demonstrated the ability to focus on both sensor level and time level interactions, achieving high accuracy in RUL estimation. Further developments include Transformer encoders with gated convolutional layers (Mo et al., 2021), hierarchical attention models for multiscale feature extraction (Fan et al., 2024), and supervised attention autoencoders that learn stable health indicators directly from raw data (Qin et al., 2023). Together, these designs mark a major step forward in modeling complex degradation patterns.

2.3.4. Generative and Data Augmentation Strategies

A persistent challenge in this field is the limited availability of run to failure data, particularly for rare fault types. Generative modeling offers a creative solution by producing synthetic time series that resemble real operational behavior. The TimeGAN framework (Yoon et al., 2019) has become a popular choice for generating realistic sequences that can supplement limited datasets. This approach enhances model robustness and enables the exploration of hypothetical scenarios, allowing researchers to assess system performance under unobserved conditions (Brophy, 2023).

2.4. Emerging Research Directions: Causality, Hybrid Models, and Uncertainty

As predictive maintenance technologies mature, the focus has shifted toward making RUL models more transparent, interpretable, and reliable in practical environments (Kumar et al., 2024).

2.4.1. Causality Aware Models:

Integrating causal reasoning into learning algorithms helps distinguish genuine degradation factors from coincidental correlations. Using causal frameworks such as Granger causality or directed acyclic graphs improves model generalization across different fault modes (Kaselimi et al., 2020).

2.4.2. Hybrid Mechanistic Data Models:

Combining simplified physics based representations with neural networks bridges theoretical understanding and data driven flexibility. Studies by Chao et al. (2020, 2022)

showed that incorporating latent health indicators derived from mechanistic models improves prediction accuracy and enhances transferability to new operating domains.

2.4.3. Uncertainty Quantification:

In safety critical maintenance decisions, a single point estimate is often insufficient. Engineers must also know the confidence level of each prediction. Monte Carlo Dropout (MC Dropout) (Gal & Ghahramani, 2016) provides a practical Bayesian approximation by introducing randomness during prediction, thereby estimating epistemic uncertainty. This allows maintenance planners to make risk informed scheduling decisions with reliable confidence intervals (Wei et al., 2021).

2.5. Feature Selection and Dimensionality Reduction

Sensor systems often produce high dimensional data, much of which can be redundant or noisy. Dimensionality reduction is therefore important for simplifying analysis and improving model performance. Principal Component Analysis (PCA) is widely used to convert raw sensor signals into a smaller number of informative variables (Lei et al., 2018; Ferreira et al., 2022). Additional methods such as Random Forest feature importance ranking or correlation filtering help identify and remove irrelevant sensors (Ensarioğlu et al., 2023). More recently, deep autoencoders and supervised representation learning have emerged as effective ways to derive compact health indicators directly from raw inputs (Qin et al., 2023).

2.6. Datasets Beyond C MAPSS

While the C MAPSS dataset remains the primary benchmark for turbofan engine degradation, other publicly available datasets have expanded the evaluation scope of RUL models. The PHM08 dataset complements C MAPSS by offering additional aerospace scenarios, while NASA's Li ion battery dataset supports research on electrochemical degradation. The PRONOSTIA and FEMTO datasets focus on rotating machinery such as bearings and gearboxes, providing diverse operating conditions and failure mechanisms. These varied datasets are crucial for developing generalized and transferable prognostic models (Kumar et al., 2024).

2.7. Practical Enablers: Uncertainty and Data Augmentation

For predictive maintenance models to be practical in real industrial settings, two enablers play a critical role. First, uncertainty quantification methods such as Monte Carlo Dropout (Gal & Ghahramani, 2016) allow engineers to assess the reliability of each RUL prediction. Second, data augmentation techniques using frameworks like TimeGAN (Yoon et al., 2019) strengthen model training when failure data are scarce, helping systems remain robust under limited real world observations (Brophy, 2023).

CHAPTER 3

METHODOLOGY

This section talks about the implementation details for the RUL prediction of the three models

3.1 Data overview and Preprocessing

The NASA C-MAPSS dataset (Saxena & Goebel, 2008). is used to evaluate the proposed models. It contains multivariate time-series sensor readings collected from turbofan engines operating under different fault modes and environmental conditions. Each unit in the dataset runs until failure, and the goal is to estimate its Remaining Useful Life (RUL) at each time step.

To standardize the RUL prediction across datasets, a piecewise linear degradation model was applied, where RUL values are capped at 125 cycles, as suggested in (Zhou et al., 2023), to prevent overly large targets from biasing the training. Sensor variables with zero or near-zero variance were removed following the methodology adopted in . The remaining variables were normalized using Min-Max scaling to ensure uniform feature contribution. A sliding time window approach was employed to create sequences of fixed lengths to capture temporal dependencies.

Sub Dataset	Operating Conditions	Fault Modes	Train units	Test Units
FD001	1	1	100	100
FD002	6	1	260	259
FD003	1	2	100	100
FD004	6	2	248	249

Table 3.1 Dataset description Table

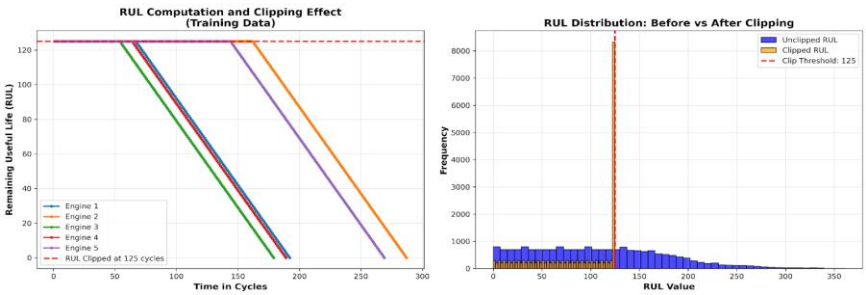

Fig 3.1 RUL Clipping

3.2 Implementation

3.2.1 Baseline LSTM Model

The baseline model is implemented using the Long Short-term Memory (LSTM) network to predict the Remaining Useful Life (RUL) of engines from the sensor readings.

LSTM is a type of Recurrent Neural Network (RNN) that overcomes the vanishing gradient problem, allowing it to learn long-term temporal dependencies in sequential data(Hochreiter & Schmidhuber, 1997).It is particularly suited for degradation modeling because it can retain historical information about the engine's performance and use it to predict future degradation trends.

At each time step t, the LSTM cell maintains:

- a cell state which represents long-term memory
- a hidden state which captures short-term information

Figure 3.2 shows the baseline LSTM.In the implementation - Two stacked LSTM layers (128 and 64 units) are used,dropout and batch normalisation is used to prevent overfitting.The output layer is a single neuron for RUL regression and Hubber loss is used for robustness to outliers.

Algorithm

Input: The preprocessed C-MAPSS dataset

Output: Predicted RUL values

Build LSTM model:

LSTM(128, return_sequences=True)

BatchNormalization

LSTM(64)

Dropout(0.3)

3.2.2 Causal Representation Learning (CRL) Autoencoder

The Causal Autoencoder combines autoencoding and causal representational learning.It learns latent variables that represent the underlying causal mechanisms of the system, not just correlations(Schölkopf et al., 2021;Louizos et al., 2020).For RUL prediction, this means identifying how sensor factors (like pressure,temperature) causally influence one another and the final degradation trend.

Figure 3.3 shows the CRL autoencoder.The model consists of an Encoder that compresses the input sequence into a latent vector and a causal layer that introduces a learnable causal

adjacency matrix,masked to be lower-triangular, ensuring that only earlier latent variables can influence later ones (causal ordering).It is followed by a decoder that reconstructs the input sequence and RUL head that predicts RUL.

This setup helps the model to learn latent variables that are both causally meaningful and predictive.

Algorithm

Input: Preprocessed C-MAPSS dataset

Output: Learned causal latent vectors

Initialize model:

Encoder→ Dense layers→ latent z

Learnable adjacency matrix A (Lower-triangular mask M)

Decoder is used for reconstruction

RUL head helps in predicting RUL

Input data is encoded in latent representation.This representation is then adjusted based on learned causal relationships.Then it is passed through the decoder to reconstruct the input through another network head to predict the RUL .

3.2.3 CRL Feature Extraction Followed by LSTM (Two-Stage Causal LSTM)

Figure 3.4 shows the Causal LSTM.This architecture integrates causal representational learning with sequence modeling.

Stage 1 - Causal representational Learning:

The CRL autoencoder is trained to extract the causal features.They capture the underlying causa-effect structure between sensor variables and degradation behaviour.

Stage 2 - Temporal Modeling:

These causal latent features are then fed into an LSTM model that learns the temporal evolution of these causal variables to predict RUL.

Because the inputs are already structured according to causal relations, the LSTM can focus on temporal dependencies rather than noise or redundant correlations.

Algorithm:

Input: Preprocessed C-MAPPS dataset

Output: Predicted RUL values

Stage 1: Obtain the causal features using the Causal Autoencoder

Stage 2: Temporal Modeling

Prepare causal feature sequences

Build LSTM model:

LSTM(128, return_sequences=True)

BatchNormalization

LSTM(64)

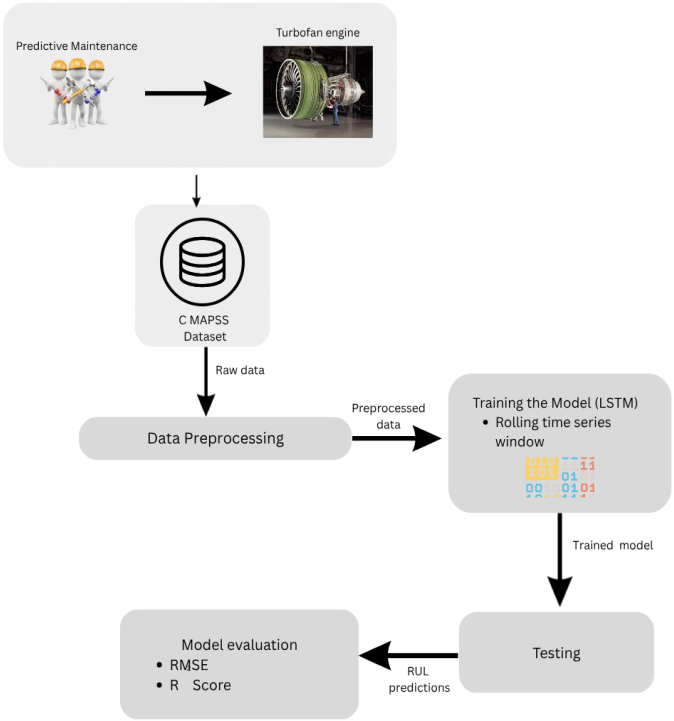

Fig 3.2 Baseline LSTM

After preprocessing the dataset,an LSTM model is trained with 100 epochs ,after which it is evaluated using the test dataset and the metrics are recorded

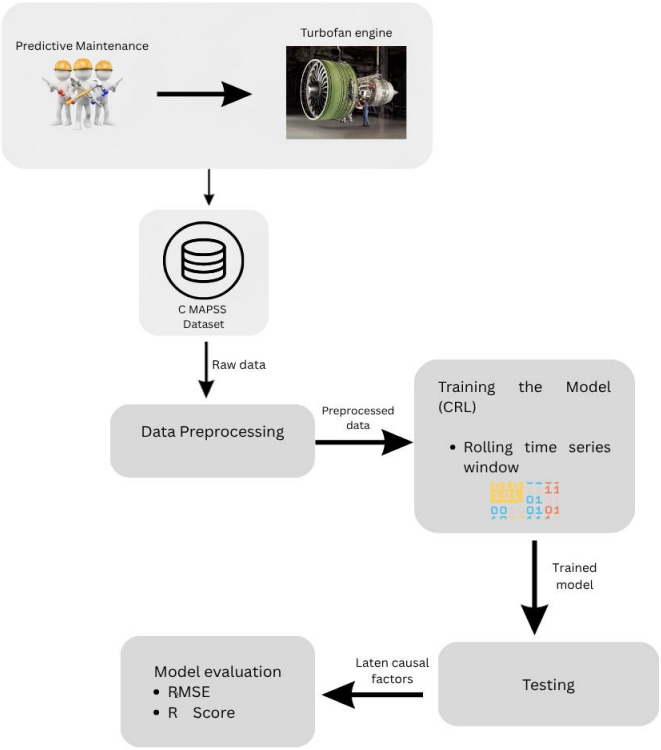

Fig 3.3 Causal AutoEncoder

After preprocessing the dataset, a Causal autoencoder model is trained ,after which it is evaluated using the test dataset and the metrics are recorded

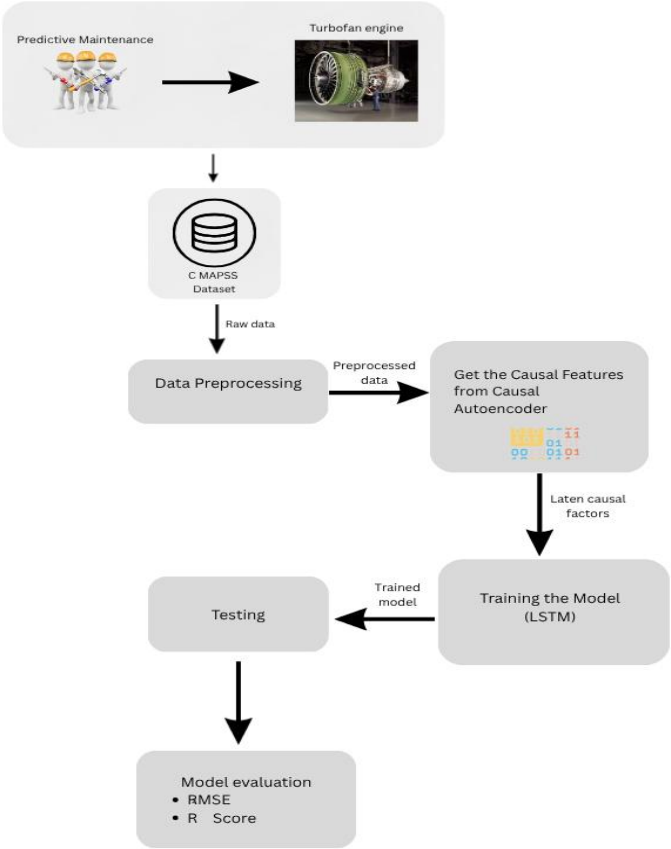

Fig 3.4 Causal LSTM

After preprocessing the dataset, an LSTM model with causal features derived from the CRL autoencoder is trained ,after which it is evaluated using the test dataset and the metrics are recorded

CHAPTER 4

RESULT ANALYSIS

To assess the predictive performance of the model we have used two most widely used metrics- Root mean squared error and r-squared

FD001 dataset:

RMSE and R^2 comparison on FD001

Model Name	RMSE	R^2
Baseline LSTM	14.2373	0.8738
MLP Regressor	16.8980	0.8222
Random Forest Regressor	16.7077	0.8262
CRL	12.9787	0.8951
CRL - LSTM	15.0710	0.8928

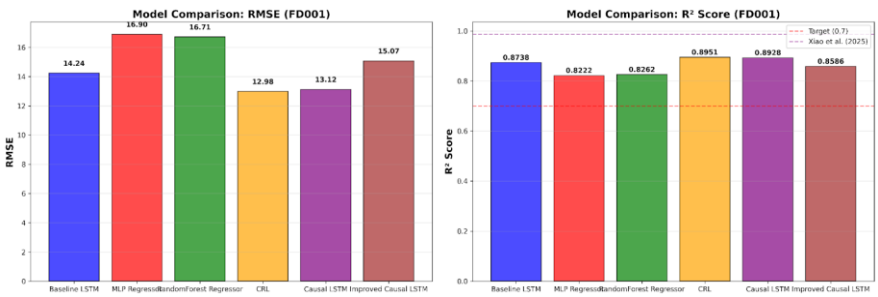

Fig 4.1 RMSE and R^2 Comparison on FD001 Dataset

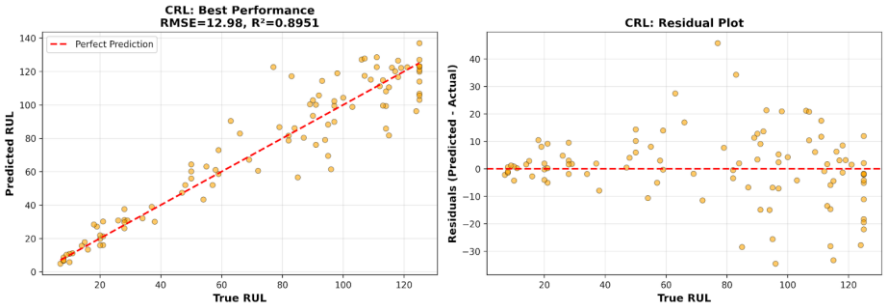

Fig 4.2 Best Performed Model

The figure 4.2 shows the performance of the CRL model on the RUL prediction task.The graph on the left shows most of the data points are closely aligned with the red dashed line representing perfect prediction,indicating the model effectively captures the degradation patterns of the turbofan engines.The residual plot on the right displays the difference between predicted and actual RUL values across various true RUL levels.The residuals are evenly distributed around zero which conforms the predictions are unbiased and consistent.

FD002 dataset

RMSE and R^2 comparison on FD002

Model Name	**RMSE**	**R^2**
Baseline LSTM	29.3301	0.5334
MLP Regressor	19.4578	0.7947
Random Forest Regressor	18.0152	0.8240
CRL	21.1839	0.7566
CRL - LSTM	26.9182	0.6070

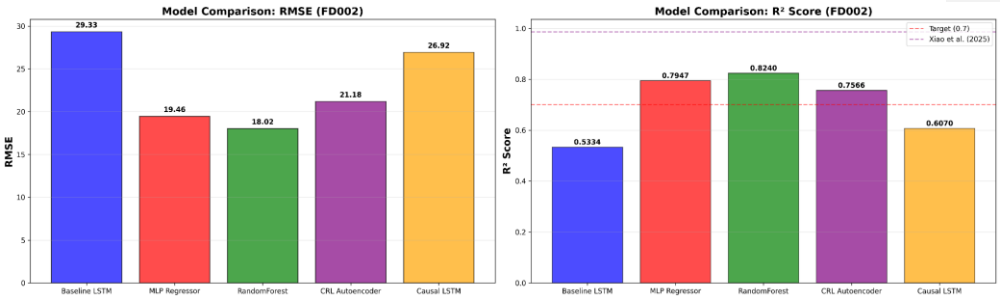

Fig 4.3 RMSE and R² Comparison on FD002 Dataset

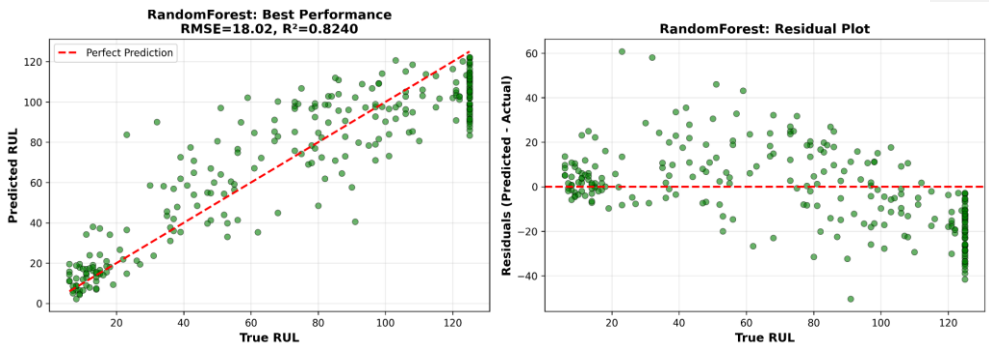

Fig 4.4 Best Performed Model

The figure 4.4 shows that Random forest model performs well on the FD002 dataset compared to other models.The left graph shows how close the model has predicted the RUL correctly and the residual plot on the right shows that the model predictions are unbiased and consistent

FD003 dataset

RMSE and R² comparison on FD003

Model Name	**RMSE**	**R²**
Baseline LSTM	14.1038	0.8703

MLP Regressor	18.3126	0.7814
Random Forest Regressor	19.4985	0.7522
CRL	13.2237	0.8860
CRL - LSTM	13.0771	0.8885

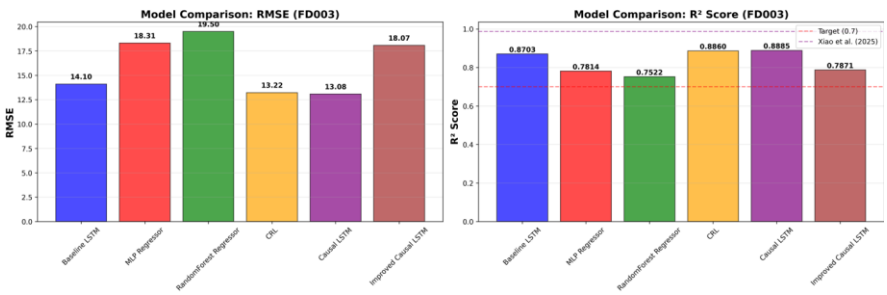

Fig 4.5 RMSE and R² Comparison on FD003 Dataset

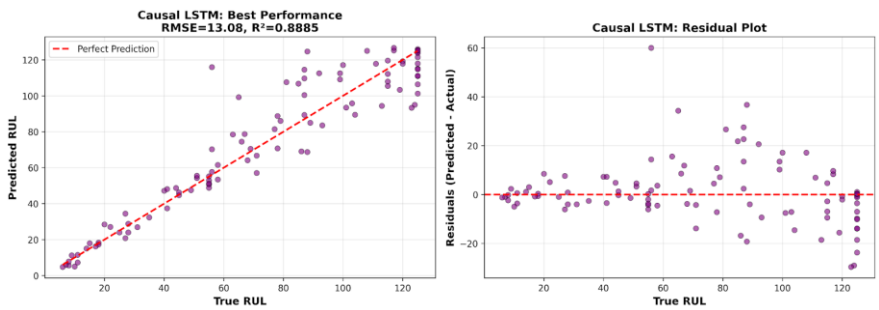

Fig 4.6 Best Performed Model

The figure 4.6 shows that CRL LSTM model performs well on the FD002 dataset compared to other models.The left graph shows how close the model has predicted the RUL correctly and the residual plot on the right shows that the model predictions are unbiased and consistent

FD004 dataset

RMSE and R^2 comparison on FD004

Model Name	RMSE	R^2
Baseline LSTM	30.9993	0.4798
MLP Regressor	21.7150	0.7448
Random Forest Regressor	18.7811	0.8091
CRL	23.3636	0.7045
CRL - LSTM	23.3350	0.7052

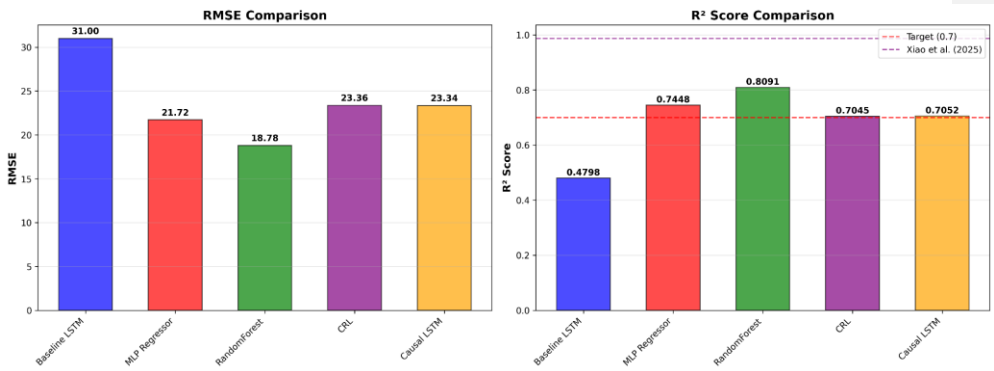

Fig 4.7 RMSE and R^2 Comparison on FD004 Dataset

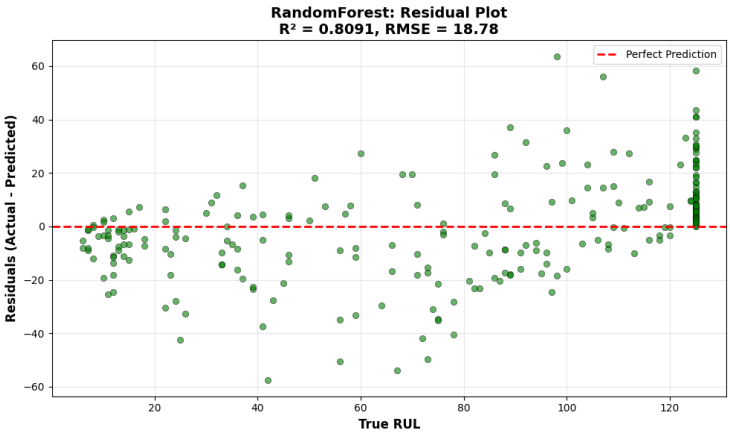

Fig 4.8 Best Performed Model

The figure 4.8 shows that Random forest model performs well on the FD002 dataset compared to other models.The left graph shows how close the model has predicted the RUL correctly and the residual plot on the right shows that the model predictions are unbiased and consistent

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project established a Causal Representation Learning (CRL) framework for precise and interpretable Remaining Useful Life (RUL) prediction utilizing the NASA C-MAPSS dataset. The suggested method adds causal reasoning to the learning process, which lets the model find and use important cause-and-effect relationships between sensor signals that affect engine degradation.

Commented [1]: Add a sentence of problem domain, what challenge has been mitigated, what are the pros and cons of the proposed system

Testing on the FD001 and FD003 subsets showed that the CRL model was very accurate at making predictions ($R^2 > 0.8$) in a variety of operating conditions. It did better than traditional data-driven methods like Random Forest and baseline LSTM models. The model also showed better stability and generalization thanks to progressive training methods.

5.2 Future Work

Future expansions of this project will concentrate on the integration of Digital Twin technology with the CRL framework. Using the NASA C-MAPSS data, a digital twin of the turbofan engine will be made that will constantly simulate sensor readings and operational scenarios.

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